Proving Data Structures Correct: Abstract Data Types

Computer-Assisted Theorem Proving (7CCSACTP)

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Using ADTs

The purpose of an ADT is to provide a context
for implementing generic algorithms
parameterised with the interface functions of the ADT.

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Example

locale `Set =`
fixes ...
assumes ...

begin
fun `set_of_list` **where**
```
"set_of_list [] = empty" |
"set_of_list (x # xs) = insert x (set_of_list xs)"
```

lemma `"invar(set_of_list xs)"`
by`(induction xs)`
```
  (auto simp:  invar_empty invar_insert)
```
end